

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

Student Name

Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file <FirstLast>_A07_GLMs.Rmd (replacing <FirstLast> with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1 set up your session
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
library(lubridate)
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
here() #check wd
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
#import raw NTL-LTER data for chem/physics
ntl_ltr <- read.csv(here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),
                    stringsAsFactors = TRUE)
```

```
# Set date to date format
class(ntl_ltr$sampleddate) #factor
```

```
## [1] "factor"
```

```
ntl_ltr$sampleddate <- as.Date(ntl_ltr$sampleddate, format = "%m/%d/%y")
```

```
#2 set the theme
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "right")
theme_set(mytheme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: Mean lake temperature in July does not change with depth across all lakes Ha: Mean lake temperature in July changes with depth across all lakes
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4 wrangle the data
#dates in july
ntl_ltr$month <- month(ntl_ltr$sampleddate)
ntl_july <- ntl_ltr %>% filter(month == 7)

#filter columns
ntl_sub <- ntl_july %>%
```

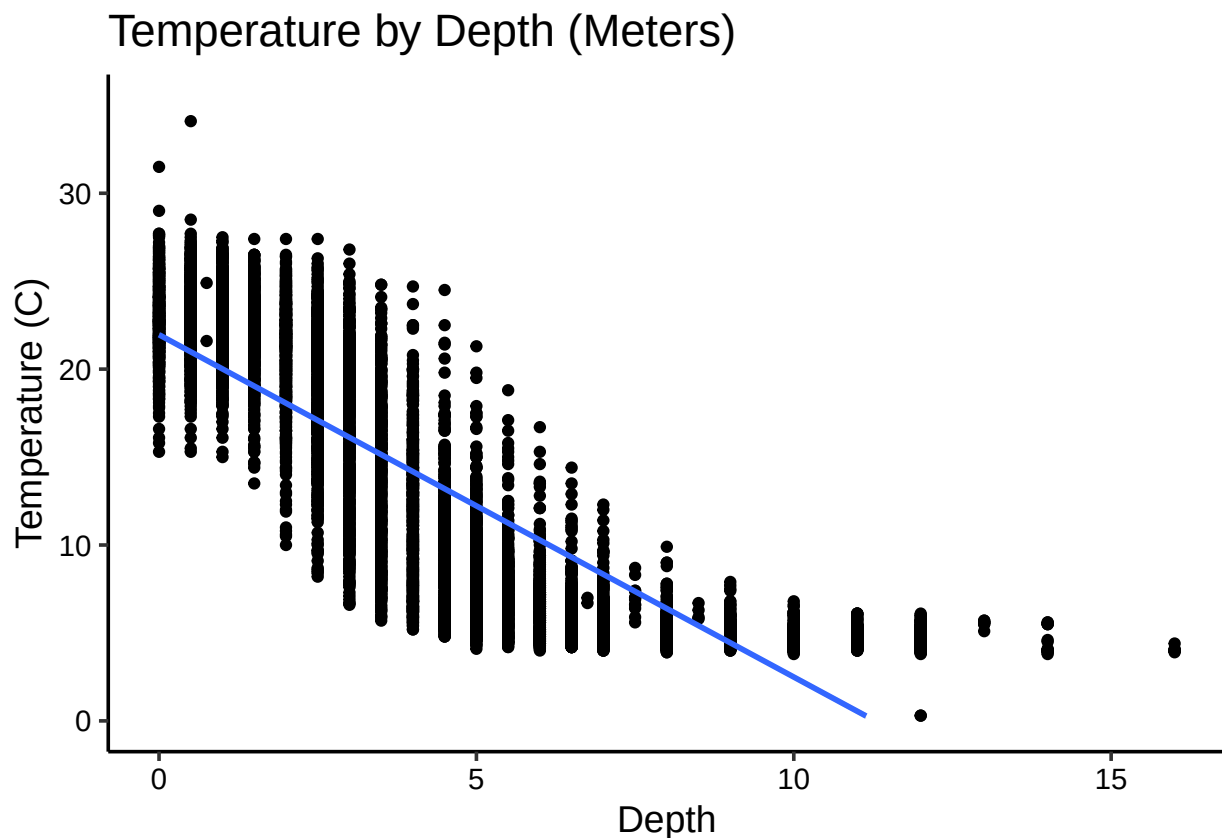
```
select(lakename, year4, daynum, depth, temperature_C) %>%
na.omit() #remove nas
```

#5 visualize the data

```
ggplot(ntl_sub, aes(x=depth, y=temperature_C)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(title = "Temperature by Depth (Meters)",
       x = "Depth", y = "Temperature (C)") +
  ylim(0, 35) +
  theme(mytheme)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: There is a negative association between the two variables. As the depth increases, the temperature decreases. The distribution of points has high variability and suggests the data may not be linear.

7. Perform a linear regression to test the relationship and display the results.

```
#7
regression <- lm(data=ntl_sub, temperature_C ~ depth)
summary(regression) #r2 = 0.7387, p < 2.2e-16

##
## Call:
## lm(formula = temperature_C ~ depth, data = ntl_sub)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: 73.87% of the variability in temperature can be explained by changes in depth. There are 9726 degrees of freedom and the p value is $< 2.2e-16$, indicating that the results are statistically significant. For every 1 meter change in depth, the temperature is predicted to decrease by 1.9 degrees celcius.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9 run an AIC to select variables
ntl_AIC <- lm(data=ntl_sub, temperature_C ~ year4 + daynum + depth)
step(ntl_AIC) #Step-wise regression
```

```
## Start: AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                141687 26066
## - year4      1         101 141788 26070
## - daynum     1         1237 142924 26148
## - depth      1      404475 546161 39189

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = ntl_sub)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##   -8.57556      0.01134      0.03978     -1.94644
```

```
#10 multiple regression
ntl_model <- lm(data = ntl_sub, temperature_C ~ year4 + daynum + depth)
summary(ntl_model) #r2 increased to 0.7412
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = ntl_sub)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994   0.32044
## year4        0.011345   0.004299   2.639   0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: Adding year4 and daynum increases the R2 slightly to 0.7412.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12 run an anova
#as aov
ntl_anova <- aov(data = ntl_sub, temperature_C ~ lakenname)
summary(ntl_anova) #p<2e-16

##              Df Sum Sq Mean Sq F value Pr(>F)
## lakenname      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#as lm
ntl_anova2 <- lm(data = ntl_sub, temperature_C ~ lakenname)
summary(ntl_anova2) #p<2.2e-16

##
## Call:
## lm(formula = temperature_C ~ lakenname, data = ntl_sub)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.174 < 2e-16 ***
## lakennameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakennameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakennameHummingbird Lake -6.8931     0.9429  -7.311 2.87e-13 ***
## lakennamePaul Lake        -3.8522     0.6656  -5.788 7.36e-09 ***
## lakennamePeter Lake       -4.3501     0.6645  -6.547 6.17e-11 ***
## lakennameTuesday Lake    -6.5972     0.6769  -9.746 < 2e-16 ***
## lakennameWard Lake        -3.2078     0.9429  -3.402 0.000672 ***
## lakennameWest Long Lake   -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

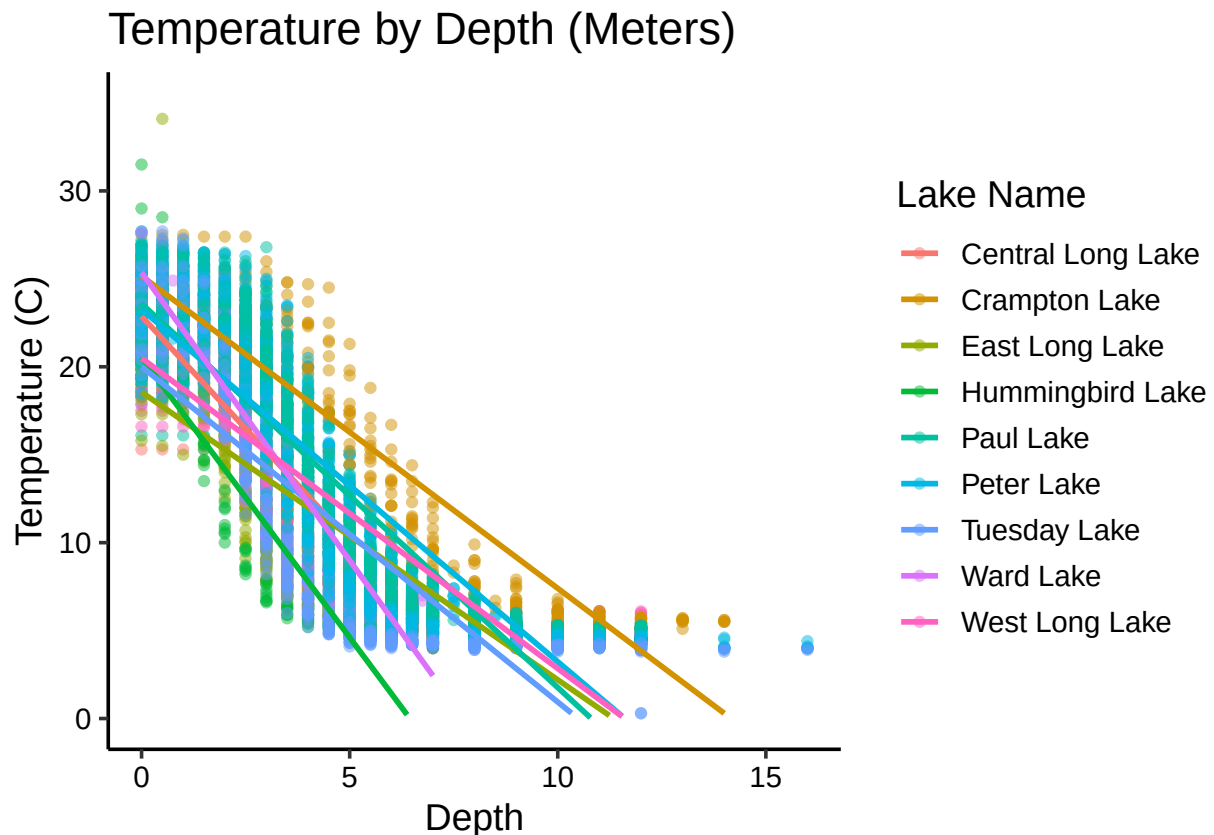
Answer: The p-value is $p < 2.2e-16$, so we can reject the null hypothesis and accept the alternative hypothesis that there is a significant difference in mean temperature among lakes.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
ggplot(ntl_sub, aes(x=depth, y=temperature_C, color=lakename)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se=FALSE) +
  labs(title = "Temperature by Depth (Meters)",
       x = "Depth", y = "Temperature (C)", color="Lake Name") +
  ylim(0, 35) +
  theme(mytheme)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
TukeyHSD(ntl_anova)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = ntl_sub)
##
## $lakename
##
```

	diff	lwr	upr	p adj
Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000
Paul Lake-Central Long Lake	-3.8521506	-5.9170942	-1.7872070	0.0000003
Peter Lake-Central Long Lake	-4.3501458	-6.4115874	-2.2887042	0.0000000
Tuesday Lake-Central Long Lake	-6.5971805	-8.6971605	-4.4972005	0.0000000
Ward Lake-Central Long Lake	-3.2077856	-6.1330730	-0.2824982	0.0193405
West Long Lake-Central Long Lake	-6.0877513	-8.2268550	-3.9486475	0.0000000
East Long Lake-Crampton Lake	-5.0842215	-6.5591700	-3.6092730	0.0000000
Hummingbird Lake-Crampton Lake	-4.5786109	-7.0538088	-2.1034131	0.0000004
Paul Lake-Crampton Lake	-1.5376312	-2.8916215	-0.1836408	0.0127491
Peter Lake-Crampton Lake	-2.0356263	-3.3842699	-0.6869828	0.0000999
Tuesday Lake-Crampton Lake	-4.2826611	-5.6895065	-2.8758157	0.0000000
Ward Lake-Crampton Lake	-0.8932661	-3.3684639	1.5819317	0.9714459
West Long Lake-Crampton Lake	-3.7732318	-5.2378351	-2.3086285	0.0000000
Hummingbird Lake-East Long Lake	0.5056106	-1.7364925	2.7477137	0.9988050
Paul Lake-East Long Lake	3.5465903	2.6900206	4.4031601	0.0000000
Peter Lake-East Long Lake	3.0485952	2.2005025	3.8966879	0.0000000
Tuesday Lake-East Long Lake	0.8015604	-0.1363286	1.7394495	0.1657485
Ward Lake-East Long Lake	4.1909554	1.9488523	6.4330585	0.0000002
West Long Lake-East Long Lake	1.3109897	0.2885003	2.3334791	0.0022805
Paul Lake-Hummingbird Lake	3.0409798	0.8765299	5.2054296	0.0004495
Peter Lake-Hummingbird Lake	2.5429846	0.3818755	4.7040937	0.0080666
Tuesday Lake-Hummingbird Lake	0.2959499	-1.9019508	2.4938505	0.9999752
Ward Lake-Hummingbird Lake	3.6853448	0.6889874	6.6817022	0.0043297
West Long Lake-Hummingbird Lake	0.8053791	-1.4299320	3.0406903	0.9717297
Peter Lake-Paul Lake	-0.4979952	-1.1120620	0.1160717	0.2241586
Tuesday Lake-Paul Lake	-2.7450299	-3.4781416	-2.0119182	0.0000000
Ward Lake-Paul Lake	0.6443651	-1.5200848	2.8088149	0.9916978
West Long Lake-Paul Lake	-2.2356007	-3.0742314	-1.3969699	0.0000000
Tuesday Lake-Peter Lake	-2.2470347	-2.9702236	-1.5238458	0.0000000
Ward Lake-Peter Lake	1.1423602	-1.0187489	3.3034693	0.7827037
West Long Lake-Peter Lake	-1.7376055	-2.5675759	-0.9076350	0.0000000
Ward Lake-Tuesday Lake	3.3893950	1.1914943	5.5872956	0.0000609
West Long Lake-Tuesday Lake	0.5094292	-0.4121051	1.4309636	0.7374387
West Long Lake-Ward Lake	-2.8799657	-5.1152769	-0.6446546	0.0021080

```
ntl_groups <- HSD.test(ntl_anova,"lakename", group = TRUE) #groups w/ same mean
ntl_groups
```

```
## $statistics
## MSerror Df Mean CV
## 54.1016 9719 12.72087 57.82135
##
## $parameters
## test name.t ntr StudentizedRange alpha
## Tukey lakename 9 4.387504 0.05
```

```
##
## $means
##           temperature_C      std      r      se Min  Max    Q25   Q50
## Central Long Lake      17.66641 4.196292  128 0.6501298 8.9 26.8 14.400 18.40
## Crampton Lake          15.35189 7.244773  318 0.4124692 5.0 27.5  7.525 16.90
## East Long Lake         10.26767 6.766804  968 0.2364108 4.2 34.1  4.975  6.50
## Hummingbird Lake       10.77328 7.017845  116 0.6829298 4.0 31.5  5.200  7.00
## Paul Lake              13.81426 7.296928 2660 0.1426147 4.7 27.7  6.500 12.40
## Peter Lake             13.31626 7.669758 2872 0.1372501 4.0 27.0  5.600 11.40
## Tuesday Lake           11.06923 7.698687 1524 0.1884137 0.3 27.7  4.400  6.80
## Ward Lake              14.45862 7.409079  116 0.6829298 5.7 27.6  7.200 12.55
## West Long Lake         11.57865 6.980789 1026 0.2296314 4.0 25.7  5.400  8.00
##
##           Q75
## Central Long Lake 21.000
## Crampton Lake    22.300
## East Long Lake   15.925
## Hummingbird Lake 15.625
## Paul Lake        21.400
## Peter Lake        21.500
## Tuesday Lake     19.400
## Ward Lake        23.200
## West Long Lake   18.800
##
## $comparison
## NULL
##
## $groups
##           temperature_C groups
## Central Long Lake      17.66641      a
## Crampton Lake          15.35189     ab
## Ward Lake              14.45862     bc
## Paul Lake              13.81426      c
## Peter Lake             13.31626      c
## West Long Lake         11.57865      d
## Tuesday Lake           11.06923     de
## Hummingbird Lake       10.77328     de
## East Long Lake         10.26767      e
##
## attr(,"class")
## [1] "group"
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Peter, Paul, and Ward lake have the same mean temperature statistically. No lakes have a temperature that is statistically unique.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: If we are looking at the difference between two lakes we could use a 2-sample t-test

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does

the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
#filter crampton and ward lake from july data
ntl_ttest <- ntl_sub %>%
  filter(lakename%in% c("Crampton Lake", "Ward Lake"))

#two-sample t-test - is temp diff b/n campton and ward?
temp.twosample <- t.test(ntl_ttest$temperature_C ~ ntl_ttest$lakename)
temp.twosample #p<0.2649 fail to reject null
```

```
##
## Welch Two Sample t-test
##
## data: ntl_ttest$temperature_C by ntl_ttest$lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129 2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##           15.35189                14.45862
```

Answer: We failed to reject the null hypothesis, meaning there is no significant difference between the mean temperatures of Ward and Crampton Lake. This matches the answer in part 16 that the lakes have the same mean temperature statistically.